

INSTRUCTIONS

1. Once you are done with your code. Push your code to your **github account**
2. Make the repository **public**. Write a Readme for installation/running or any other information
3. Test if it's accessible publicly.
4. Share the repository link in the google form provided

PROBLEM STATEMENT

You are building the backend for a **smart flashcard system**. Users can add flashcards with just a question and answer — no subject is specified by them. Your backend will automatically determine the subject of the flashcard based on the text (e.g., presence of keywords like "photosynthesis" → "Biology").

Later, a student can request flashcards from any subject, and your API should return a mixed batch intelligently.

Task 1: Add Flashcard with Subject Inference

Endpoint:

POST /flashcard

```
{  
  
  "student_id": "stu001",  
  
  "question": "What is Newton's Second Law?",  
  
  "answer": "Force equals mass times acceleration"  
}
```

Output:

```
{  
  
  "message": "Flashcard added successfully",  
}
```

```
"subject": "Physics"
}
```

Use any method that you want for classification logic:

Option 1 Can be Rule-based :

Option 2 Any Pre-trained model or trained model:

If you are training the model, you need to figure out the data

Requirements:

- Detect the subject of the flashcard (e.g., "Physics") based on the question.
- Store student_id, question, answer, and subject.

Task 2: Get Flashcards by Mixed Subjects

Endpoint:

```
GET /get-subject?student_id=stu001&limit=5
```

Output:.

```
[
  {
    "question": "What is Newton's Second Law?",
    "answer": "Force equals mass times acceleration",
    "subject": "Physics"
  },
  {
    "question": "What is photosynthesis?",
    "answer": "A process used by plants to convert light into energy",
    "subject": "Biology"
  }
]
```

```
}  
]
```

Behavior:

- Return up to limit number of flashcards from different subjects.
- Shuffle to mix subjects.
- Only return flashcards for the given student